

Problem

In the circuit shown in Fig. 7.81

$$v(t) = 56e^{-200t} \text{ V}, \quad t > 0$$

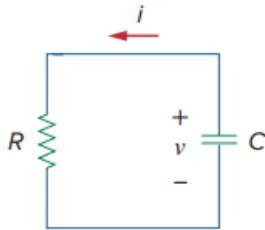
$$i(t) = 8e^{-200t} \text{ mA}, \quad t > 0$$

(a) Find the values of R and C .

(b) Calculate the time constant τ .

(c) Determine the time required for the voltage to decay half its initial value at $t = 0$.

Fig. 7.81:



Step-by-step solution

Step 1 of 5

Consider the voltage across the capacitor, $v(t) = 56e^{-200t} \text{ V}, t > 0$

Consider the current that flows through the circuit in Figure 1, $i(t) = 8e^{-200t} \text{ mA}, t > 0$

Consider the following circuit diagram:

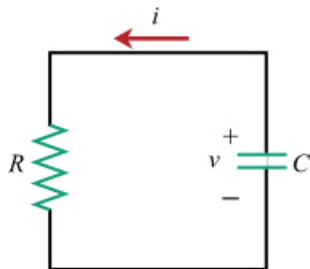


Figure 1: Source free RC circuit

Step 2 of 5

Write the generalized expression for the voltage across the capacitor for the source free RC circuit.

$$v(t) = V_0 e^{\frac{-t}{\tau}}$$

Here, V_0 represents the initial voltage at $t = 0$

τ represents the time constant and is equal to RC

Calculate the resistor value by using the Ohm's law.

$$\begin{aligned} R &= \frac{v(t)}{i(t)} \\ &= \frac{56e^{-200t}}{(8e^{-200t}) \times 10^{-3}} \\ &= \frac{56}{8 \times 10^{-3}} \\ &= 7 \text{ k}\Omega \end{aligned}$$

Thus, the value of the resistor is $7 \text{ k}\Omega$.

[Comments \(1\)](#)



Anonymous

The function $v(t)$ is kept in volts, you then made a current conversion for mA to A, thus wouldn't the final answer be ohms rather than kohms.

Step 3 of 5

Calculation of capacitor value:

Compare the generalized expression $v(t) = V_0 e^{\frac{-t}{\tau}}$ with the $v(t) = 56e^{-200t} \text{ V}$, $t > 0$ then value of the time constant which is equal to RC is,

$$\begin{aligned} \frac{1}{\tau} &= 200 \\ \frac{1}{RC} &= 200 \end{aligned}$$

Substitute the value $7 \text{ k}\Omega$ in place of R to get the value of a capacitor.

$$\begin{aligned} \frac{1}{(7 \times 10^3)C} &= 200 \\ C &= \frac{1}{(7 \times 10^3)200} \\ C &= \frac{1}{(1400 \times 10^3)} \\ C &= 0.7143 \mu\text{F} \end{aligned}$$

Thus, the value of the capacitor is $0.7143 \mu\text{F}$.

Step 4 of 5

Calculation of the time constant τ :

Compare the generalized expression $v(t) = V_0 e^{\frac{-t}{\tau}}$ with the $v(t) = 56e^{-200t}$ V, $t > 0$ then value of the time constant is,

$$\frac{1}{\tau} = 200$$

$$\tau = \frac{1}{200}$$

$$\tau = 0.005$$

$$\tau = 5 \text{ ms}$$

Thus, the value of the time constant τ is 5 ms.

Step 5 of 5

The initial voltage by comparing the generalized expression $v(t) = V_0 e^{\frac{-t}{\tau}}$ with the expression

$$v(t) = 56e^{-200t} \text{ V, } t > 0 \text{ then,}$$

$$V_0 = 56 \text{ V}$$

Determine the time required for the voltage to decay half its initial value.

$$\frac{V_0}{2} = 56e^{-200t}$$

$$\frac{56}{2} = 56e^{-200t}$$

$$e^{-200t} = \frac{1}{2}$$

Apply the logarithm on both sides of the equation.

$$-200t = \ln\left(\frac{1}{2}\right)$$

$$-200t = -0.6932$$

$$t = \frac{-0.6932}{-200}$$

$$t = 3.4657 \times 10^{-3}$$

$$t = 3.466 \text{ ms}$$

Thus, the time required for the voltage to decay half its initial value is 3.466 ms.